A database with a large amount of Multicolumn indexes could indicate excessive or redundant indexing, where single column indexes would serve the same purpose and consume less space. You'll want to look for overlapping indexes for possible optimization. We'll cover that later.

On the other hand, Multicolumn Indexes can provide some of the best performance when each of the columns covered in the definition are used by a query. Columns are used from left to right. The leftmost column in a Multicolumn Index definition can be used by queries that only need that single column. Soon you'll learn how to make sure all columns of Multicolumn Indexes are being used.

On the same PostgreSQL wiki page, use the "Index size/usage statistics" query for the Rideshare database. This query will show usage information.

To simulate some usage of Rideshare, from your Terminal run `bin/rails server` from the Rideshare directory to start the Rails Server. From another Terminal window, run `bin/rails simulate:app_activity` which will send some requests to the Rideshare server.

With that completed, you will now have some Index usage information from queries that were executing from that simulation. The number of scans shows a count of how many Index Scans have been performed using this index. The tuples fetched and tuples read show information on the number of tuples provided by this index. Indexes with no scans and no tuples fetched may be unused or infrequently used indexes.

Creating good Indexes for your database is challenging because it requires knowledge of multiple factors including the data statistics, the query patterns, and the Index types available.

Is the order of how columns appear in the Index significant?

Understanding Index Column Ordering

The order of columns covered by a Multicolumn Index is significant.

Consider a Multicolumn index that covers columns "a" and "b", ordered `(a, b)` (a then b) from the left to the right. Which types of queries can use this Index? Queries with column "a" and "b" can use this index, or queries with only column "a" can use this index. Queries with only column "b" cannot use this index. Only the first column of a Multicolumn index can be used.

This makes Multicolumn indexes a type of specialized index. They are useful for specific queries that work with multiple fields, when those fields are in